

```
rayyan@DESKTOP-UPNN1J5: ~$  
hart 1 starting  
hart 2 starting  
init: starting sh  
$ echo Rayyan <your ERD>  
missing file for redirection  
$ echo Rayyan <31604 >  
open 31604 failed  
$ echo Rayyan 31604  
Rayyan 31604  
$ pgtbltest  
print_ptbl starting  
va 0x0 pte 0x21BE4058 pa 0x86F90000 perm 0x50  
va 0x1000 pte 0x21BE4C1B pa 0x86F93000 perm 0x15  
va 0x2000 pte 0x21BE5817 pa 0x86F94000 perm 0x17  
va 0x3000 pte 0x21BE9407 pa 0x86FA5000 perm 0x7  
va 0x4000 pte 0x21BE88D7 pa 0x86FA2000 perm 0xD7  
va 0x5000 pte 0x0 pa 0x0 perm 0x0  
va 0x6000 pte 0x0 pa 0x0 perm 0x0  
va 0x7000 pte 0x0 pa 0x0 perm 0x0  
va 0x8000 pte 0x0 pa 0x0 perm 0x0  
va 0x9000 pte 0x0 pa 0x0 perm 0x0  
va 0x3FFFFFF000 pte 0x0 pa 0x0 perm 0x0  
va 0x3FFFFFF7000 pte 0x0 pa 0x0 perm 0x0  
va 0x3FFFFFF8000 pte 0x0 pa 0x0 perm 0x0  
va 0x3FFFFFF9000 pte 0x0 pa 0x0 perm 0x0  
va 0x3FFFFFFA000 pte 0x0 pa 0x0 perm 0x0  
va 0x3FFFFFFB000 pte 0x0 pa 0x0 perm 0x0  
va 0x3FFFFFFC000 pte 0x0 pa 0x0 perm 0x0  
va 0x3FFFFFFD000 pte 0x21BDF813 pa 0x86F7E000 perm 0x13  
va 0x3FFFFFFE000 pte 0x21BDFCC7 pa 0x86F7F000 perm 0xC7  
va 0x3FFFFFFF000 pte 0x20001C4B pa 0x80007000 perm 0x4B  
print_ptbl: OK  
vmprint_test starting  
page table 0x0000000086f8d000  
..0x0000000000000000: pte 0x0000000021be4051 pa 0x0000000086f91000  
..0x0000000000000000: pte 0x0000000021be4801 pa 0x0000000086f92000  
...0x0000000000000000: pte 0x0000000021be485b pa 0x0000000086f93000 RXU  
...0x0000000000000000: pte 0x0000000021be4c1b pa 0x0000000086f93000 RXU  
...0x0000000000000000: pte 0x0000000021be5817 pa 0x0000000086f94000 RMU  
...0x0000000000000000: pte 0x0000000021be9407 pa 0x0000000086fa5000 RM  
...0x0000000000000000: pte 0x0000000021be88d7 pa 0x0000000086fa2000 RMU  
..0x0000003fc0000000: pte 0x0000000021be3801 pa 0x0000000086f8e000  
..0x0000003ffff00000: pte 0x0000000021be3c01 pa 0x0000000086f8f000  
...0x0000003ffffd000: pte 0x0000000021bdf813 pa 0x0000000086f7e000 RU  
...0x0000003ffffe000: pte 0x0000000021bdfcc7 pa 0x0000000086f7f000 RM  
...0x0000003fffff000: pte 0x0000000020001c4b pa 0x0000000080007000 RX  
vmprint_test: done  
pgaccess_test starting  
pgaccess_test: OK  
ugetpid_test starting  
ugetpid_test: OK  
ksuper_test: OK  
pgtbltest: all tests succeeded  
$
```

```
$ QEMU: Terminated  
rayyan@DESKTOP-UPNN1J5: ~/xv6-labs-2026$ ./grade-lab-pgtbl  
make: 'kernel/kernel' is up to date.  
== Test pgtbltest == (1.7s)  
== Test pgtbltest: vmprint_test ==  
pgtbltest: vmprint_test: OK  
== Test pgtbltest: pgaccess ==  
pgtbltest: pgaccess: OK  
== Test pgtbltest: ugetpid ==  
pgtbltest: ugetpid: OK  
== Test pgtbltest: superpg ==  
pgtbltest: superpg: OK  
== Test answers-pgtbl.txt ==  
answers-pgtbl.txt: OK  
== Test usertests ==
```